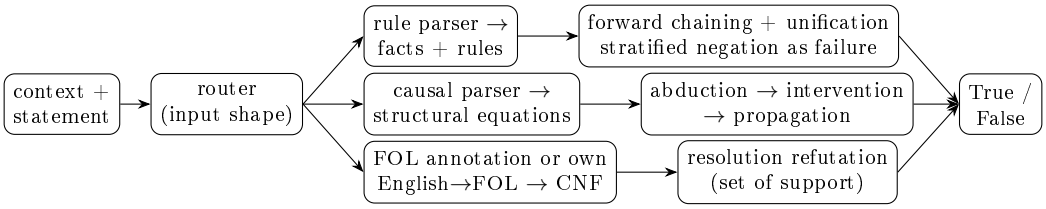
AI Assignment: A Classical Reasoning Agent for Logical Entailment

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1 Overview

The agent receives a knowledge base in English (*context*) and a *statement*, and outputs True if the statement is entailed and False otherwise. Conversion to an internal representation is done by a hand-written, template based parser; every decision is taken by a classical reasoning engine. No learned or generative model is used at any stage.



The router tries, in order: (1) the causal engine for "Would Y occur ..." questions over "...causes ..." text; (2) resolution when an FOL version of the premises is supplied; (3) forward chaining when at least 90% of the context sentences fit the rule templates (the closed-world assumption is only safe when the context was understood); (4) our own English→FOL translation followed by resolution for other English; (5) otherwise False. The binary label means *entailed* → True; *contradicted or unknown* → False.

2 Representation

Literals and rules (RuleTaker-style text). A literal is `(predicate, arguments, negated)`; arguments are constants (`bob`, `bald_eagle`) or the variable `?x` produced by *someone / something / they / it*. Adjectives become unary predicates and verbs binary ones after normalising to the base form:

Sentence	Internal form
The cat does not chase the cow.	$\neg\, chase(cat, cow)$
If someone is young and not kind then they like the dog.	$young(x) \wedge \neg kind(x) \Rightarrow like(x, dog)$
All red, round things are not big.	$red(x) \wedge round(x) \Rightarrow \neg big(x)$

About ten templates cover facts, *if–then* rules (with or without *then*), *all / plural* group rules and *someone who is ...* rules. Clauses without a subject ("...and smart") inherit the previous subject. On the fixed 10,000-query RuleTaker sample used for the ablations, all 146,351 context sentences matched a template (this measures parsing coverage, not semantic correctness).

Structural equations (CounterBench). Each sentence "A and B together cause not C" becomes the boolean equation $C := \neg(A \wedge B)$; "or" gives a disjunction. Variables with no causes are exogenous. "We observed V" is evidence, "$X \sim \mathrm{Bern}(p)$" a prior, and the question gives the intervention (e.g. "if not X instead of X" $\Rightarrow$ $\mathrm{do}(X{=}0)$ with real-world evidence $X{=}1$).

Clauses (FOLIO). FOLIO provides an FOL version of every premise. A recursive-descent parser handles $\forall, \exists, \neg, \wedge, \vee, \rightarrow, \leftrightarrow, \oplus$ and repairs small annotation slips (unbalanced brackets, "$x{=}y$").

English→FOL without annotation. For free-form English (FOLIO premises without FOL) we translate the patterns we recognise into FOL strings: *All/Every A (who B) are C* $\Rightarrow \forall x(A \wedge B \rightarrow C)$, *No A is B*, *Some A are B / There is a A*, *If ..., then ...*, *X is A or B*, *X is neither A nor B*, generic plurals (*Penguins are flightless birds*) and facts about named individuals. A property phrase becomes one unary predicate after lower-casing, dropping articles and singularising (*is a German writer* $\Rightarrow$ $german_writer(x) \wedge writer(x)$). Unsupported sentences are skipped, so True is returned only if the query follows from the understood sentences, and not when the translated premises also prove its negation.

From FOL to clauses. Formulas are converted to CNF: eliminate $\rightarrow, \leftrightarrow, \oplus$; push $\neg$ inward; rename variables; Skolemise $\exists$; distribute $\vee$ over $\wedge$.

3 Reasoning algorithms

Forward chaining with unification. Rule conditions are matched against the fact base by depth-first search with unification, and new conclusions are added until a fixed point is reached. RuleTaker is built under the

closed-world assumption (Table 3: 99.94% closed vs. 75.11% open world), so by default $\neg P$ is *negation as failure*: it holds when P cannot be derived. This is an explicit setting (`CLOSED_WORLD`); set to `False`, $\neg P$ must be stated or derived, i.e. ordinary open-world entailment. To make NAF sound, rules are split into *strata*: a rule that tests $\neg P$ runs only after every rule able to derive P has finished. Levels are kept per (predicate, sign), so a rule that *concludes* $\neg P$ does not delay rules that use P. For a positive query the engine stops as soon as it is derived. Each derived fact stores its rule and premises, which gives the proof trace.

Counterfactual reasoning by model checking. The causal module uses the three classical steps: (i) *abduction*: enumerate every assignment of the exogenous variables (state-space search over 2^k worlds, $k \leq 3$ here) and keep those consistent with the evidence; (ii) *intervention*: overwrite the intervened variables and cut their incoming edges; (iii) *prediction*: propagate values in topological order (repeated to a fixed point if the graph has a loop). The answer is `True` only if the target holds in *every* consistent world, which is entailment by model checking. If the evidence cannot be produced by the model, the observation is held fixed as a condition instead.

Resolution refutation. To prove φ we add the clauses of $\neg\varphi$ and search for the empty clause. The search uses the *set-of-support* strategy (one parent always descends from $\neg\varphi$) with *unit preference* (shortest clause first), unification with occurs check, tautology deletion, and duplicate removal by canonical renaming. It stops after 300 steps or 0.3 s and answers `False` when no refutation is found.

4 Experiments on the public datasets

Labels are mapped to entailment: RuleTaker *entailment*→True; FOLIO True→True, False/Uncertain→False; CounterBench yes→True. No component is trained, so the public data is only used for evaluation and error analysis. Times are wall-clock times of the notebook on a laptop CPU (single thread, pure Python).

Public evaluation split	Queries	Accuracy (%)	Total time (s)	ms / query
RuleTaker test	151,911	**99.97**	132.6	0.87
FOLIO validation (with FOL annotation)	203	**84.73**	7.8	38.5
CounterBench (alpha 1,000 + backdoor 200)	1,200	**94.58**	0.48	0.40
`submission.csv` run: all public evaluation queries	153,314	**99.91**	141.3	0.92
Analysis only (training split, not held out):				
FOLIO train (with FOL annotation)	1,001	81.02	26.1	26.1

Table 1: Accuracy and wall-clock inference time. 153,314 = 151,911 + 203 + 1,200. The notebook's evaluation cell also adds FOLIO train, giving 154,315 queries at 99.79%.

RuleTaker (test)		CounterBench		FOLIO validation	
depth-0 / 1 / 2	100.00 / 99.99 / 99.99	basic	100.0	gold True	62.5
depth-3 / 3ext	99.96 / 99.96	conditional	92.0	gold False	95.2
depth-5	99.92	joint / nested	96.0 / 96.0	gold Uncertain	98.6
NatLang / 3ext-NatLang	100.00 / 99.97	backdoor	87.5		

Table 2: Accuracy (%) by reasoning depth, question type and gold label.

Variant	Acc. (%)	Time (s)	Note
RuleTaker, strata per (predicate, sign) + early stop (final)	99.94	9.05	
RuleTaker, same but full closure (no early stop)	99.94	9.21	
RuleTaker, strata by predicate name only	99.92	8.65	cycles evaluated jointly
RuleTaker, open world (no negation as failure)	75.11	8.19	data is closed-world
CounterBench nested: intervene only on first variable	78.0	–	
CounterBench nested: intervene on both (final)	96.0	–	

Table 3: Ablations. RuleTaker rows use the same fixed random sample of 10,000 test queries.

FOLIO input mode	always False	English only (our translation)	gold FOL annotation
validation (203)	64.53	66.50	84.73
train (1,001)	61.24	63.34	81.02

Table 4: FOLIO accuracy (%) by input mode. In the English-only mode the FOL fields are removed before routing.

Error analysis. The few RuleTaker errors are depth-5 theories where negation sits inside a cycle and no stratification exists. CounterBench errors are mostly backdoor and conditional items whose gold answer we could not reproduce from the stated equations under our reading of the question types, checked by hand on sampled errors (e.g. a gold answer that would require intervening on X to change X's own parent). FOLIO misses with the annotation are *True* items whose proof needs equality reasoning, very long chains, or predicates written differently in two premises (annotation typos). In the English-only mode most sentences match a pattern, but the same concept is often phrased differently in premise and query (*performs in talent shows* vs. *is a talent-show performer*), so few proofs connect and most answers stay False; the gain over the always-False baseline is therefore about two points.

5 Reasoning traces

RuleTaker, depth-5 theory, with negation as failure. Query: *The cat is kind.* ⇒ True

```
kind(cat) [by rule: If something sees the lion and it is not blue then it is kind.]
  see(cat, lion) [by rule: If something chases the cat then the cat sees the lion.]
    chase(mouse, cat) [by rule: If something is kind then it chases the cat.]
      kind(mouse) [by rule: If something sees the lion and it is not blue then it is kind.]
        see(mouse, lion) [given fact]
        ¬blue(mouse) [not derivable -> assumed false]
  ¬blue(cat) [not derivable -> assumed false]
```

CounterBench, conditional. *Ziklo causes not Blaf, Blaf and Trune together cause Vork, Vork causes Lumbo. We observed Trune. Would Lumbo occur if not Ziklo instead of Ziklo?* ⇒ True

```
equations : blaf := not ziklo ; vork := blaf and trune ; lumbo := vork
abduction : evidence {ziklo=1, trune=1} -> one consistent world {ziklo=1, trune=1}
do(ziklo=0): blaf=1, vork=1, lumbo=1 -> True in every consistent world
```

FOLIO. Premises $\forall x\,(\mathit{Square}(x) \rightarrow \mathit{FourSided}(x))$, $\forall x\,(\mathit{FourSided}(x) \rightarrow \mathit{Shape}(x))$; query $\forall x\,(\mathit{Square}(x) \rightarrow \mathit{Shape}(x))$. The negated query is Skolemised to $\mathit{square}(sk)$, $\neg\mathit{shape}(sk)$.

```
resolve [¬shape(sk)] with [shape(x) ∨¬foursided(x)] => ¬foursided(sk)
resolve [square(sk)] with [¬square(x) ∨foursided(x)] => foursided(sk)
resolve [¬foursided(sk)] with [foursided(sk)] => □(query entailed: True)
```

Same FOLIO story, English only (annotation hidden). Our translator produces $\forall x(\mathit{square}(x) \rightarrow \mathit{four_sided}(x) \wedge \mathit{sided}(x))$, $\forall x(\mathit{four_sided}(x) \wedge \mathit{sided}(x) \rightarrow \mathit{shape}(x))$ and the query $\forall x(\mathit{square}(x) \rightarrow \mathit{shape}(x))$; resolution then reaches $\square$ in five steps (True).

6 Limitations and conclusion

The English parsers are pattern based: they covered every RuleTaker and CounterBench sentence in our tests, but only part of open-domain English, so FOLIO accuracy depends strongly on whether an FOL version of the premises is available (84.7% vs. 66.5% on validation). Resolution is limited in time and steps, equality is treated as an ordinary predicate, and negation inside a rule cycle (no stratification) is handled by evaluating those rules together.

In summary, the agent combines three classical engines behind one router: stratified forward chaining for rule bases (99.97% on 151,911 RuleTaker test queries at under 1 ms per query), abduction–intervention–propagation for counterfactual questions (94.6% on CounterBench), and resolution refutation for first-order premises. Trace support is implemented for all three engines (derivation tree, consistent worlds, resolution steps) and demonstrated in Section 5; no learned or generative model is involved in any decision. The combined 99.91% is dominated by RuleTaker and should not be read as equally strong performance on unrestricted English, where FOLIO without annotations reaches 66.5%.